

Functional Requirement Document

Based on the provided Business Requirement Document (BRD), I will create a detailed Functional Requirement Document (FRD) adhering to the "Coding Requirements" section. Here are the functional requirements in bullet points:

FR-AURA-001: Parse AURA Input Payload

- Title: Parse AURA Input Payload
- Description: The system must decode, normalize, and transform AURA input payloads into a canonical XML string.
- Preconditions: I_AURA_INPUT_BASE64 is not null and connected; POLICY_NUMBER is not null and connected.
- Main Flow / Functional Steps:
 - Read I_AURA_INPUT_BASE64 and POLICY_NUMBER inputs.
 - Base64-decode I_AURA_INPUT_BASE64 and store in AURA_OUTPUT.
 - If AURA_OUTPUT starts with "{", parse as JSON and extract interviewDetails into AURA_OUTPUT.
 - Perform progressive decoding loop (max 100 attempts) until AURA_OUTPUT starts with "<".
 - If AURA_OUTPUT is XML, proceed; otherwise, log error and do not produce PARSED_AURA_OUTPUT.
 - Apply XSLT transformation based on Aura version (14 or 15) using \$\$M_XSLT_FILE or \$\$AURA15_M_XSLT_FILE.
 - Remove XML declaration encoding attribute and wrap content in "XMLA" container element.
 - Instantiate Transformer and transform wrapped XML string to produce canonical output.

FR-FILE-001: Generate 10K Flat File

- Title: Generate 10K Flat File
- Description: The system must generate a 10K flat file with derived fields and filename.
- Preconditions: Source data is available; \$\$M_SRC_SQL is configured and valid.
- Main Flow / Functional Steps:
 - Retrieve data from T_APPLICATION_STAGING and T_BENEFICIARY using \$\$M_SRC_SQL.
 - Derive Field_1..Field_n using mplt_BNCPLS_IF23_10K_Field_Calc mapplet.
 - Derive File_Path_Name using EXP_Calc_10K_FileName expression.
 - Apply Transaction Control (TC_Seperate_10K_Files) to split output into multiple files (approx. 10,000 records per file).

- Write derived fields and filename to File_Structure_10K flat file target.

FR-FILE-002: Derive Filename

- Title: Derive Filename
- Description: The system must derive a unique filename for the 10K flat file.
- Preconditions: EXP_Calc_10K_FileName expression is configured; \$\$OUT_DIR and other required parameters are set.
- Main Flow / Functional Steps:
 - Derive File_Path_Name using EXP_Calc_10K_FileName expression based on interface code, business date/batch id, sequence, and extension.
 - Ensure unique filename for parallel/batch runs by incorporating run-id or GUID.

FR-TRANS-001: Apply Transaction Control

- Title: Apply Transaction Control
- Description: The system must apply transaction control to split output into multiple files.
- Preconditions: TC_Seperate_10K_Files transformation is configured; \$\$MAX_ROWS_PER_FILE is set (default: 10,000).
- Main Flow / Functional Steps:
 - Group rows into transactions based on \$\$MAX_ROWS_PER_FILE.
 - Commit and initiate new output file context when threshold is reached.
 - Ensure each file has a unique filename.

FR-ERR-001: Handle Errors and Exceptions

- Title: Handle Errors and Exceptions
- Description: The system must handle errors and exceptions during AURA parsing, mapplet execution, and file writing.
- Preconditions: Error handling is configured; logging is enabled.
- Main Flow / Functional Steps:
 - Log errors and exceptions during AURA parsing, mapplet execution, and file writing.
 - Increment error count and allow session to continue unless threshold is exceeded.
 - Provide reject target or error file for capturing error details.